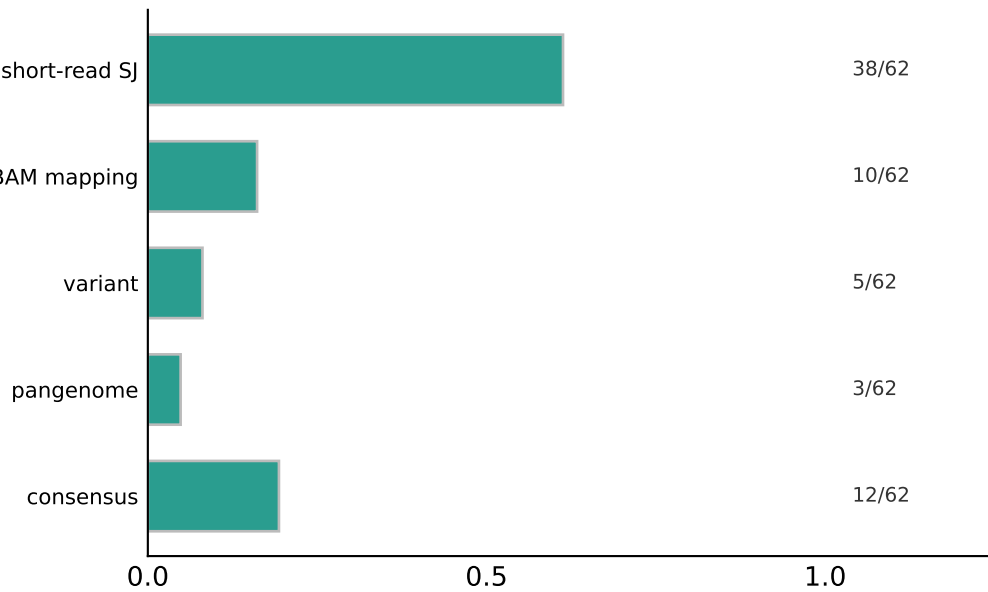


PanIsoGuard adjudication — executive summary

Orthogonal confidence layer — SQANTI3 features consumed as priors, never recomputed.

Evidence axes in play (fraction of records evaluable)



Headline — 49 novel isoforms

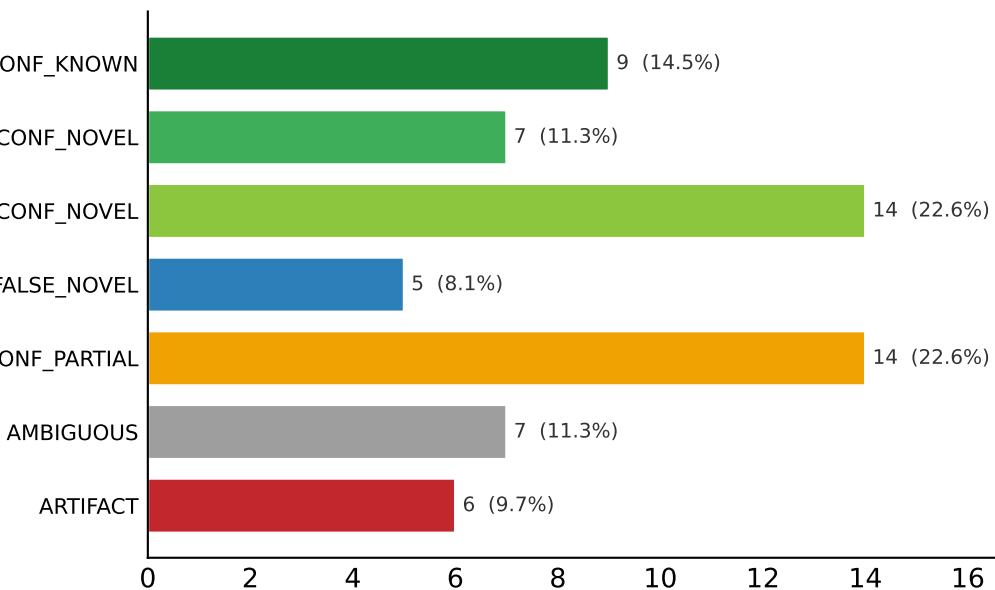
Trustworthy (HIGH/MEDIUM_CONF_NOVEL): 21

Need a look (LOW_CONF_PARTIAL/AMBIGUOUS): 17

Look like artifacts (ARTIFACT): 6

Reclassified as reference-bias (RESCUED): 5

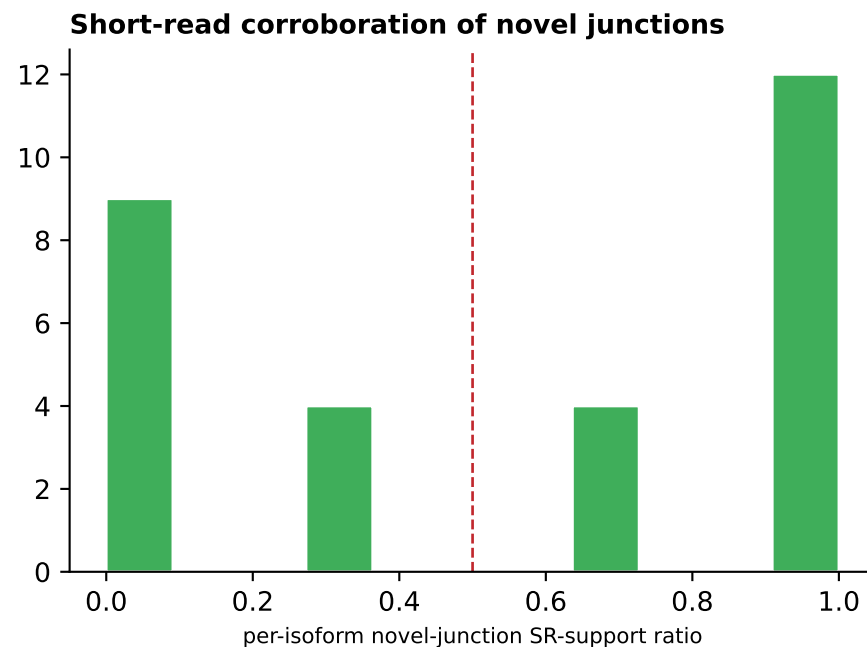
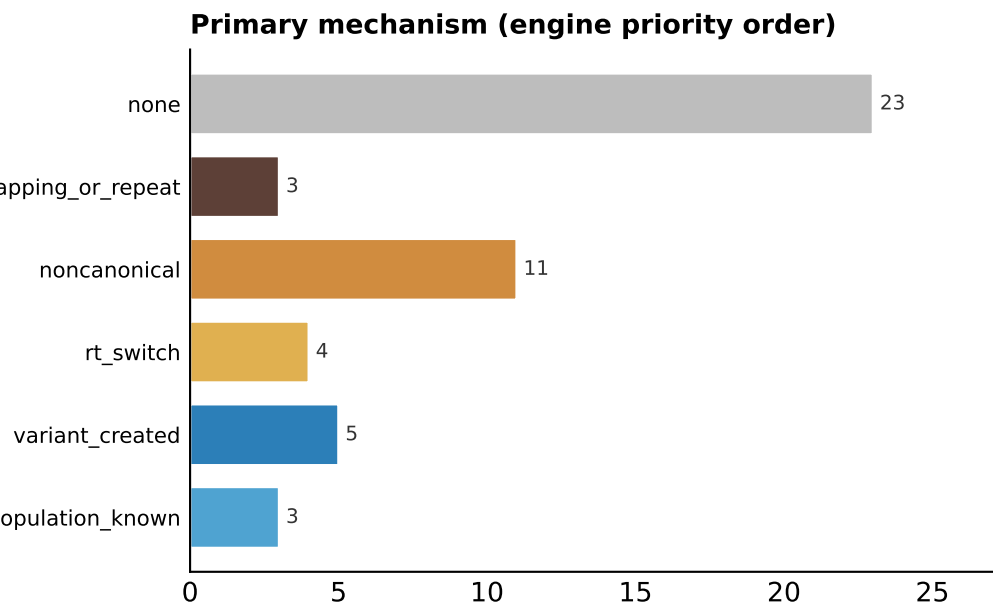
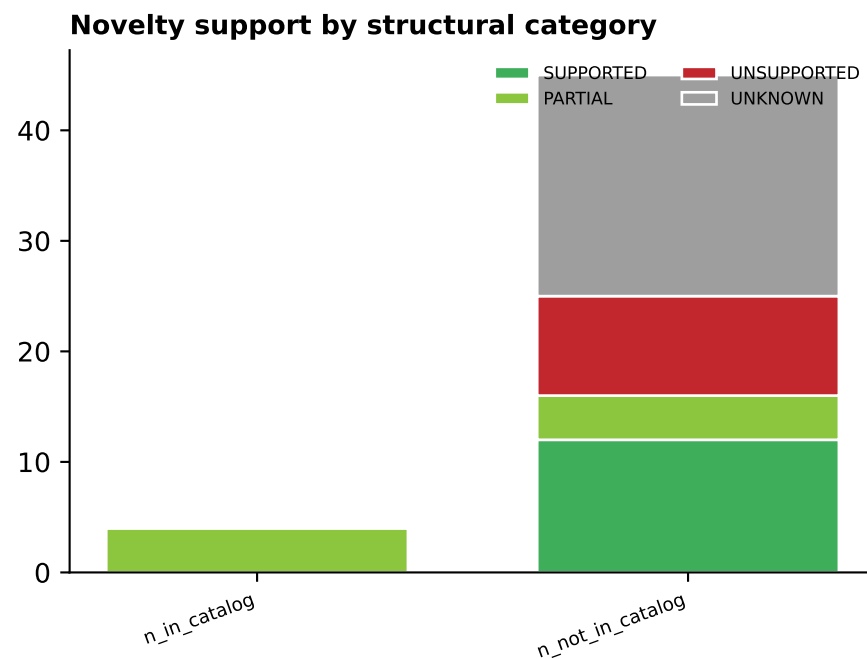
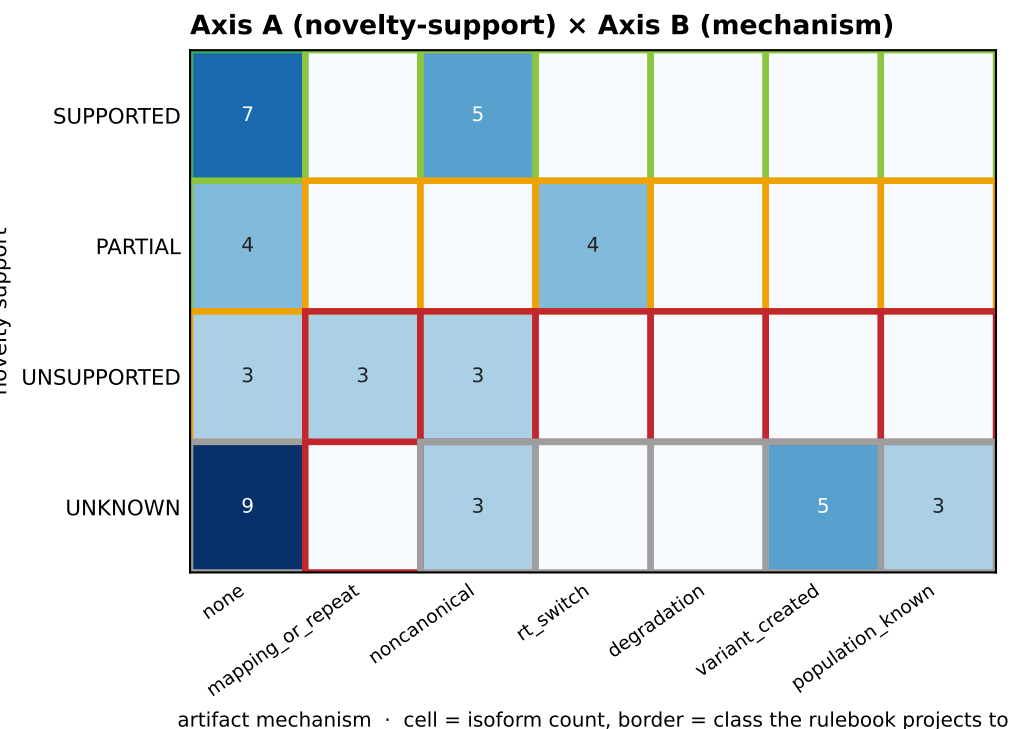
Confidence-class distribution (all isoforms)



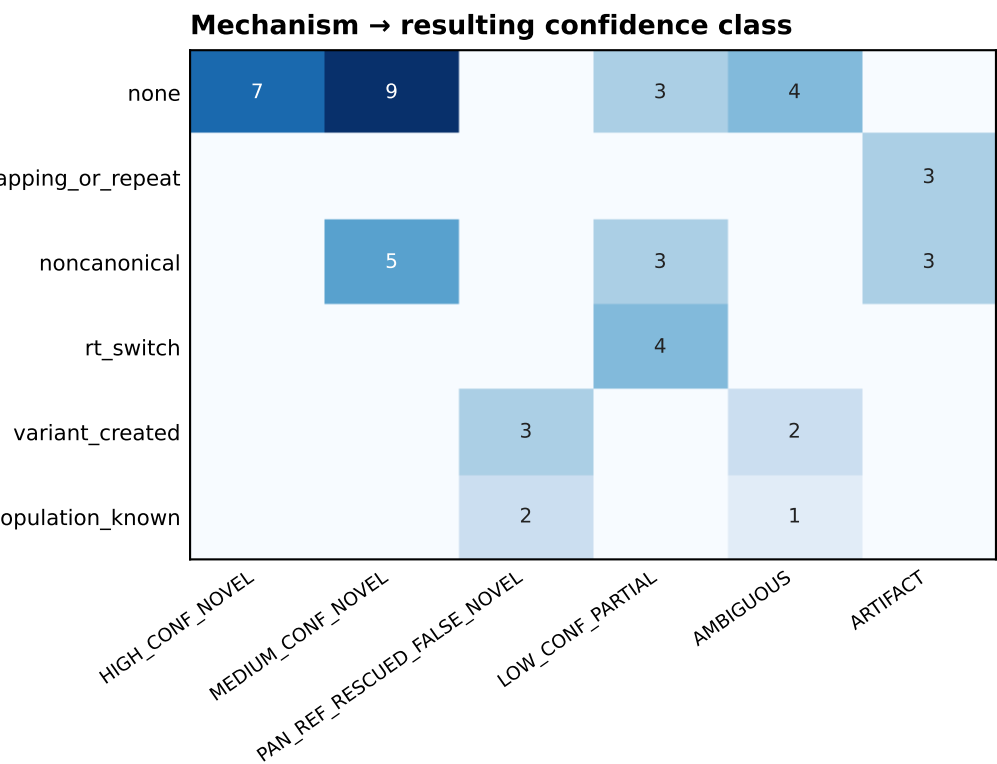
Circularity firewall — promotion conservatism



The two-axis decision space — how each verdict was reached

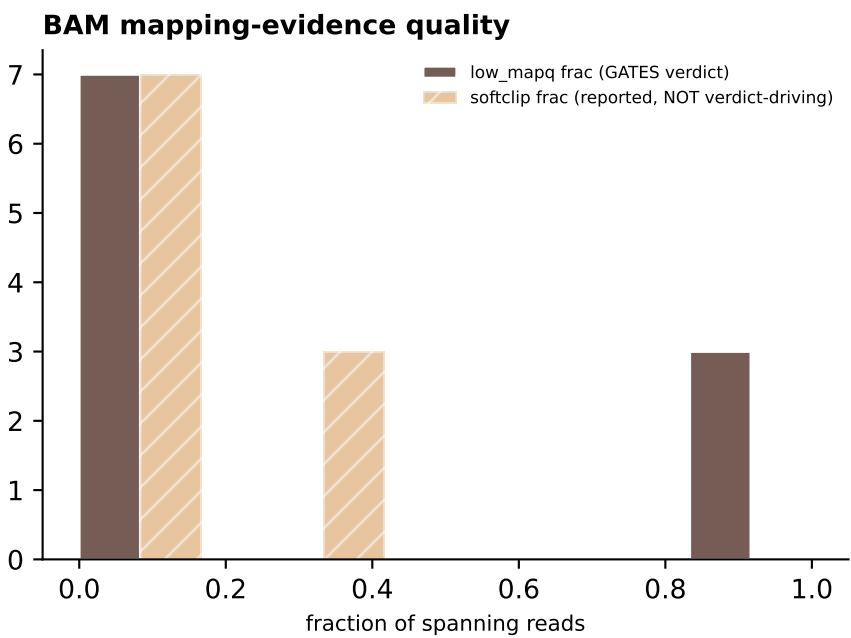


Artifact-mechanism diagnostics — why calls were down-weighted



Structural-signal availability (null = grey, never counted false)

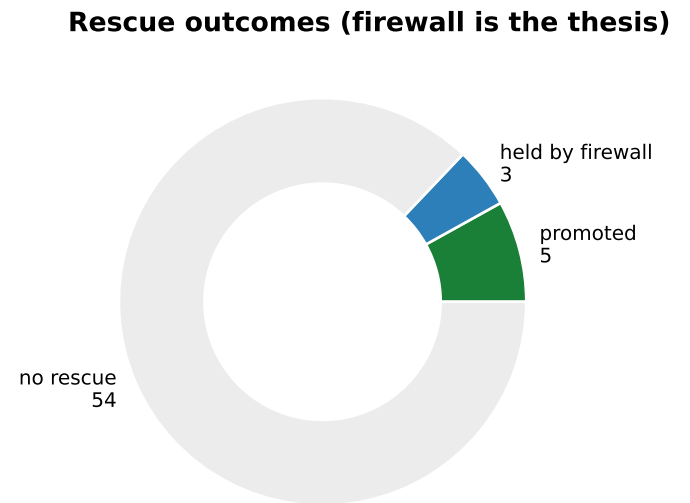
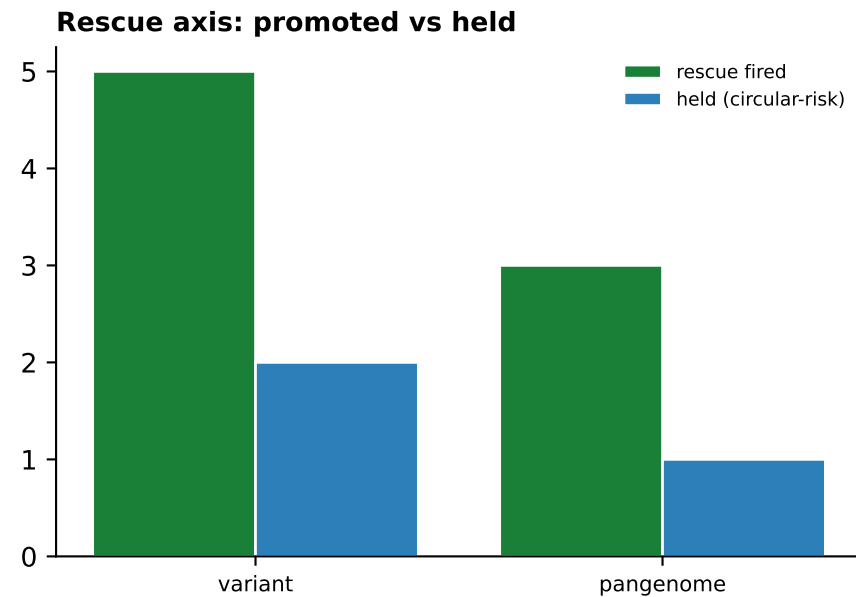
structural prior	true	false	n/a
noncanonical motif	11	51	0
RT-switch stage	4	58	0
caller chain available	62	0	0



Mapping-down-weighted calls (3)

isoform	class	n_span	low_mapq	suppl
NNC_map.36	ARTIFACT	14	0.86	0.71
NNC_map.37	ARTIFACT	14	0.86	0.71
NNC_map.38	ARTIFACT	14	0.86	0.71

Reference-bias rescue & circularity firewall — defensibility

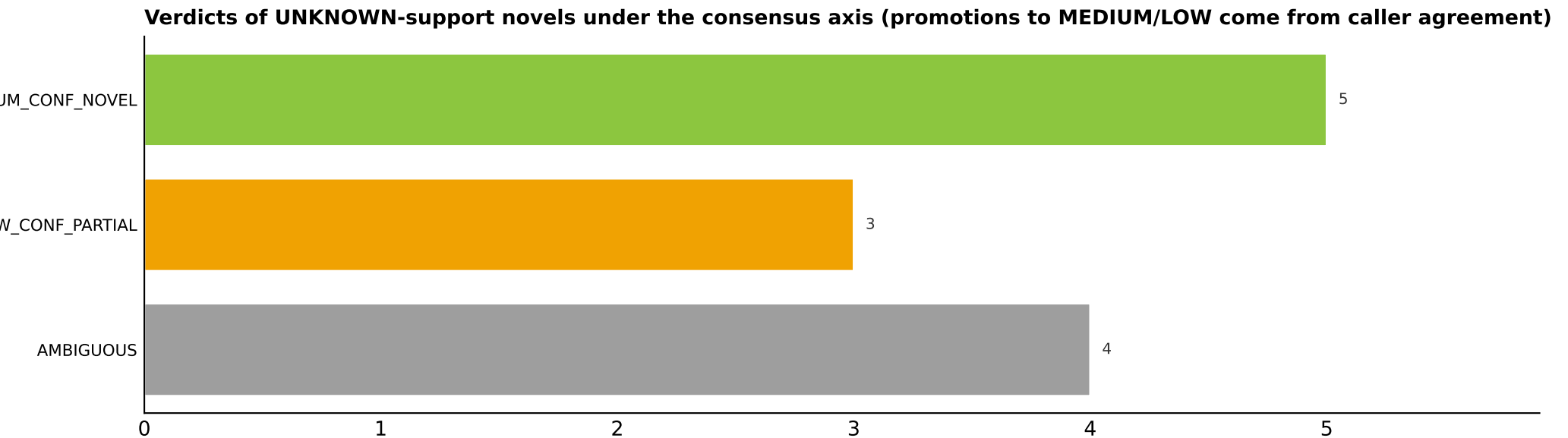
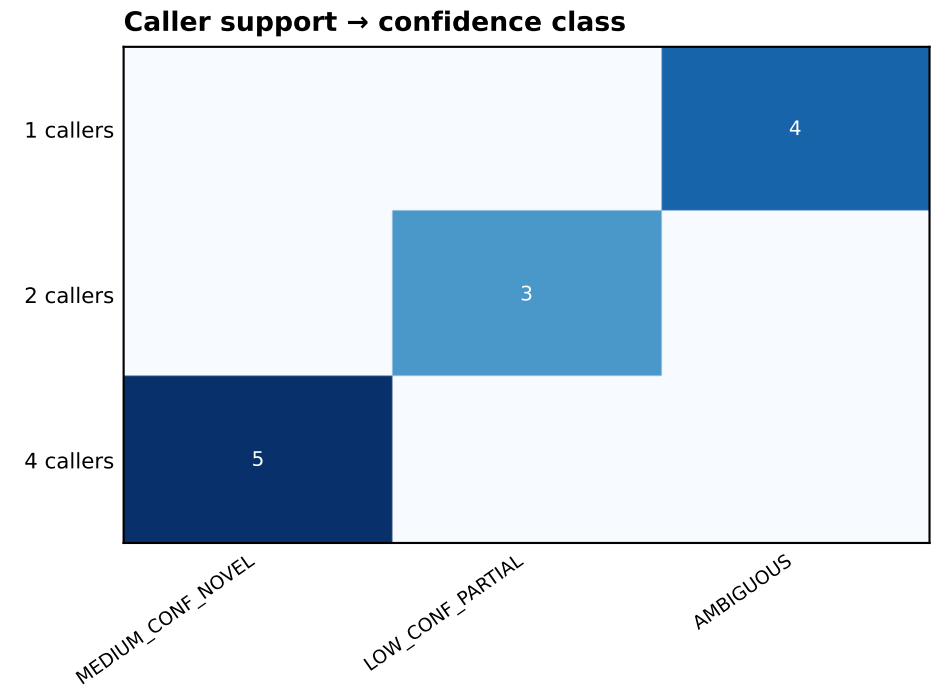
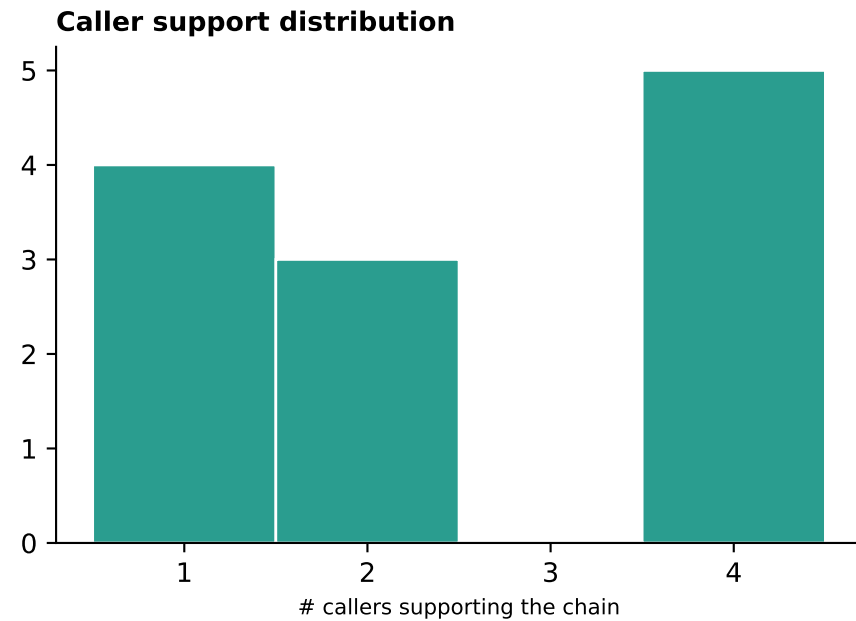


Held-rescue ledger — circularity firewall (3)

isoform	structural_category	class	why held (rule_trace)
HELD_var.44	novel_not_in_catalog	AMBIGUOUS	variant creates canonical motif on haplotype but provenance is circula
HELD_var.45	novel_not_in_catalog	AMBIGUOUS	variant creates canonical motif on haplotype but provenance is circula
HELD_pan.46	novel_not_in_catalog	AMBIGUOUS	all 2/2 novel junctions realizable on a pangenome path, but the juncti

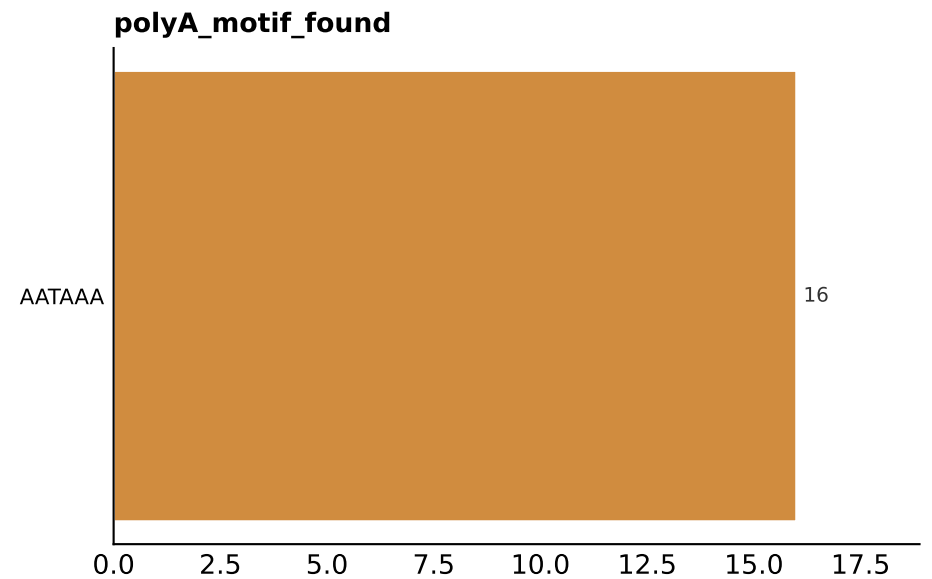
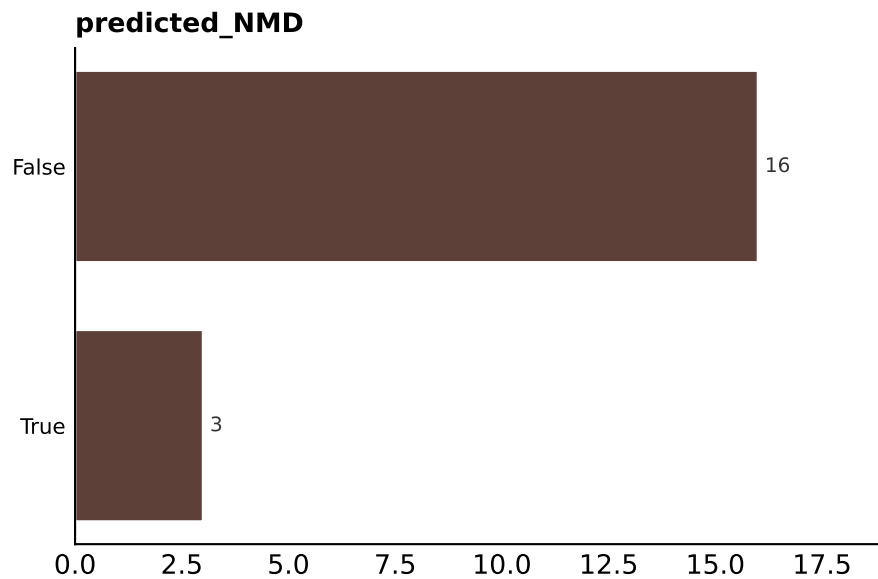
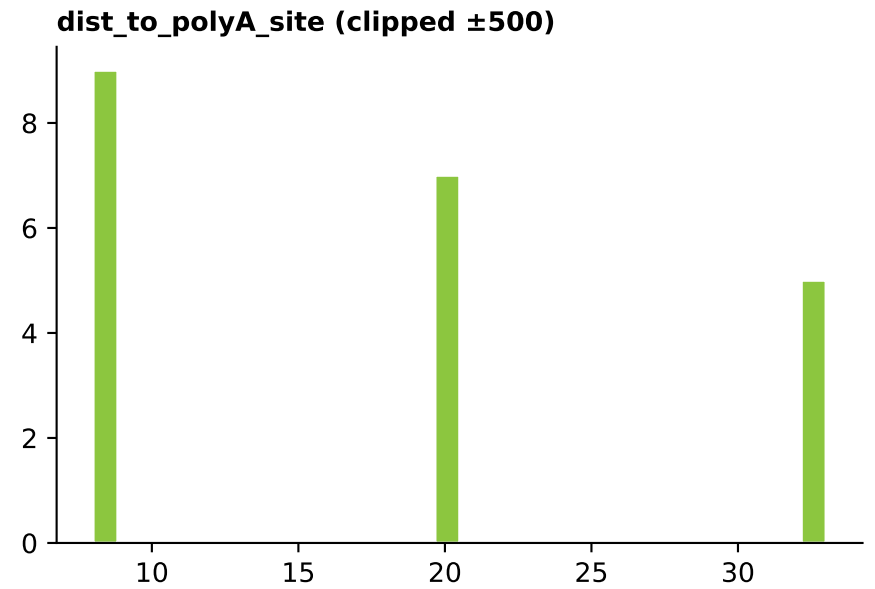
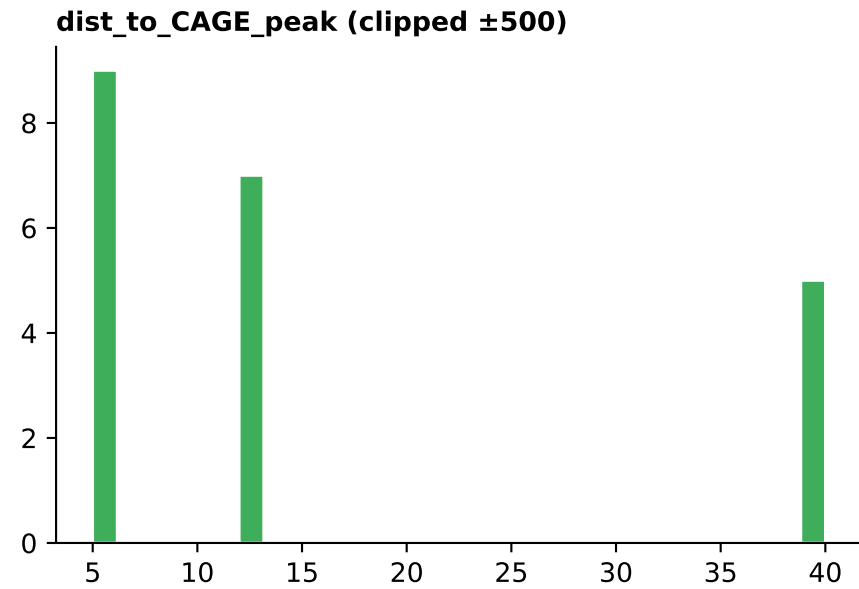
Multi-caller consensus axis

Consensus is methodological corroboration — capped at `MEDIUM_CONF_NOVEL`, never `HIGH`.



SQANTI3 biological priors (pass-through)

These are SQANTI3 annotations passed through verbatim and are NOT used in the PanIsoGuard verdict.

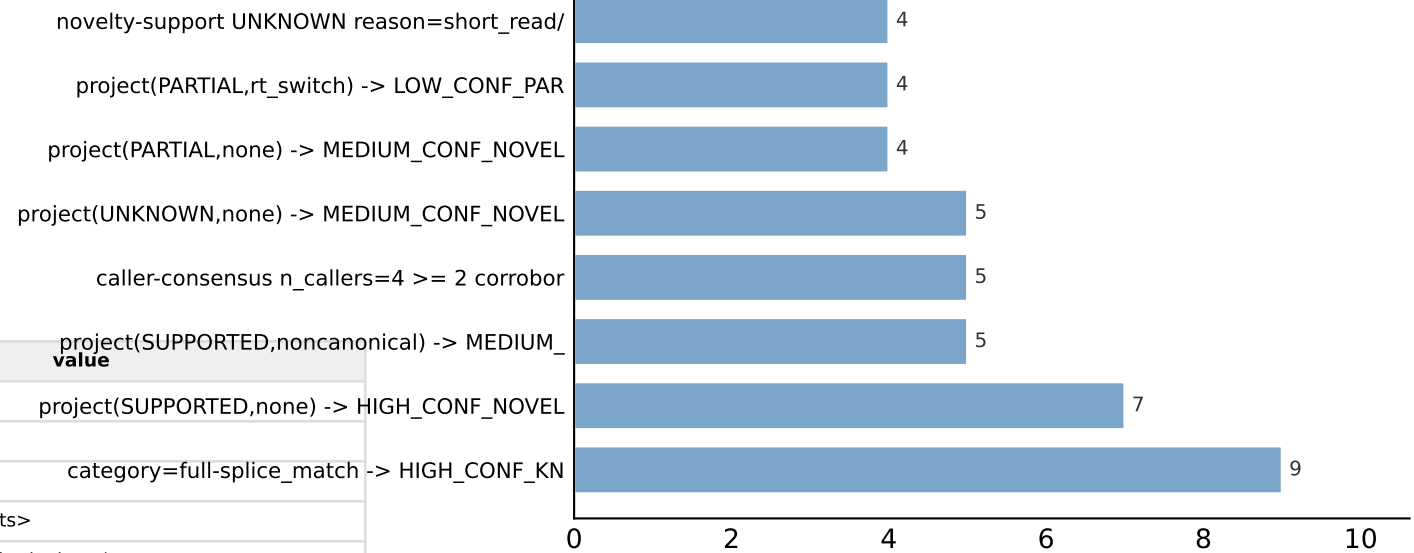


Methods, provenance & defensibility appendix

Run provenance

key	value
tool_version	0.0.3
ruleset_version	builtin-0.0.1
sqanti3_version_target	6.0
config	<built-in defaults>
classification	showcase (synthetic demo)
isoforms	showcase (synthetic demo)
ref_gtf	showcase (synthetic demo)
sj_tab	showcase (synthetic demo)
bam	showcase (synthetic demo)
axis.short_read	on
axis.catalog	on
axis.bam	on
axis.variant	on
axis.pangenome	on

Most frequent rule-trace lines



Defensibility checklist

- ✓ class tally reconciles (provenance ↔ JSONL)
- ✓ firewall held 3 circular-risk rescue(s)
- i consensus axis used this run
- i axes in play: SR, BAM, variant, pangenome, consensus
- i SQANTI3 priors consumed, never recomputed